CIE - I

Date: 26-11-24

Scheme and Solutions

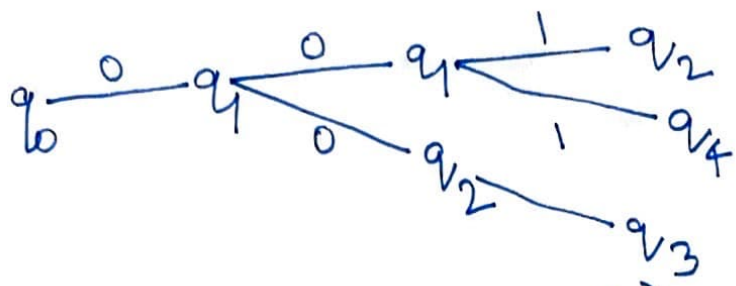
Subject : TOC

Sub. Code: CS354TA

Marks : 60

PART- A

1.0 $\delta^*(q_0, 001)$



q_4 – final state ∴ accept — 2

2.0 ϵ-closure({3, 5, 12})

$= \epsilon$-closure(3) ∪ ϵ-closure(5) ∪ ϵ-closure(12)

= {3, 6, 7, 8, 10} ∪ {5, 6, 7, 8, 10} ∪ {1, 2, 4, 12, 13} — 2

= {1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 13} — 1

3.0 i) aba or any other string — 1

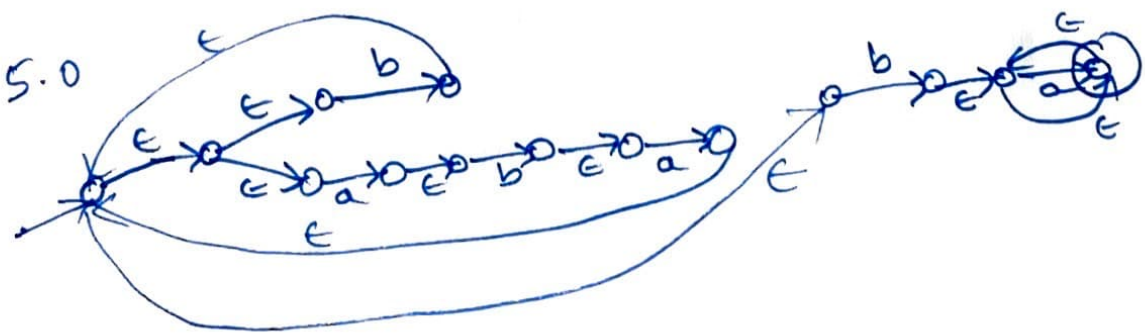
ii) b or any other string

4.0 L = {ab, ac, c}, $L \cdot L = L^2$ = {abab, abac, abc, acab, acac, acc, cab, cac, cc}

$L^2 \cdot L = L^3$ = {ababab, ababac, ababc, abacab, abacac, abacc, abcab, abcac, abcc, acabab, acabac, acabc, acacab, acabac, acacc, accab, accac, accc, cabab, cabac, cabc, cacab, cacac, cacc, ccab, ccac, ccc} — 2

$L^4 = L^3 \cdot L$ = { . . . } $|L^4| = 81$.

5.0



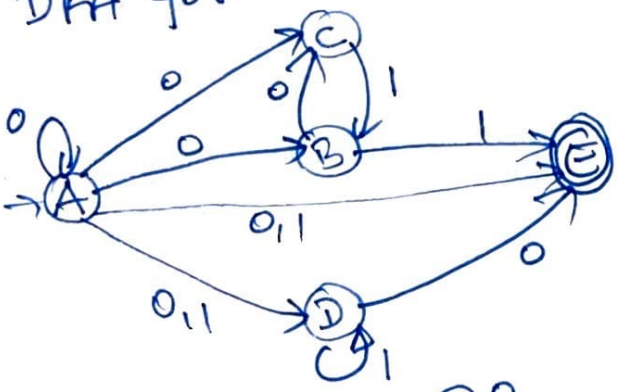
— 2

PART - B

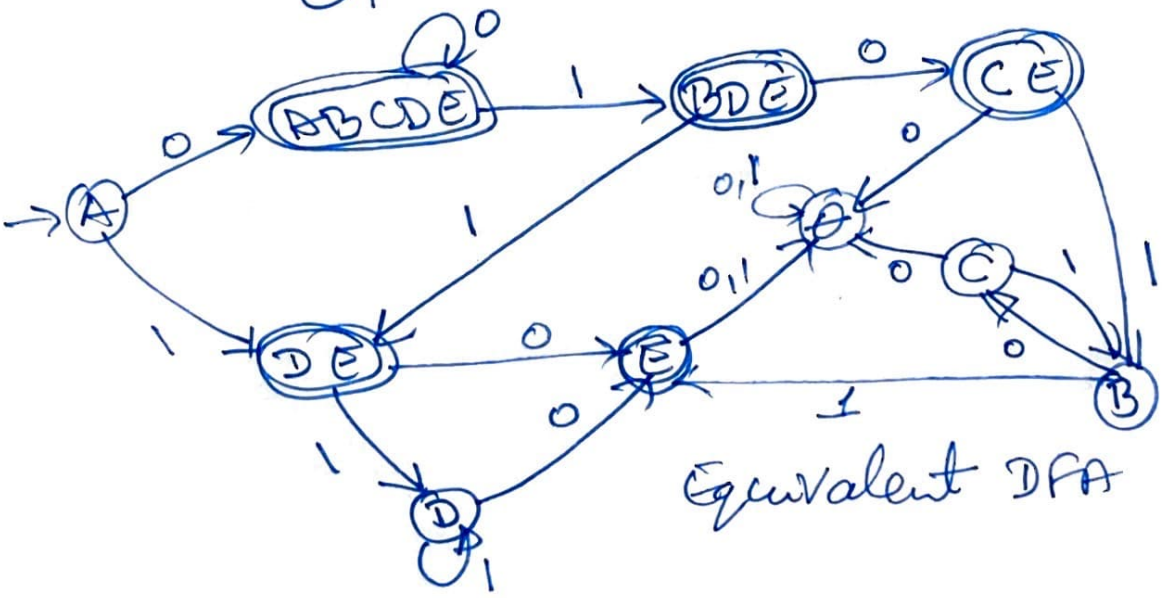
1.a Definition of regular expression — 2
R.E for (i) — 2
R.E for (ii) — 2

1.b ε-NFA definition — 1
Extended transition fn of ε-NFA — 2
Language acceptance of ε-NFA — 1

2. Equivalence of ε-NFA to NFA — 2
Equivalence of NFA to DFA — 2
Equivalent NFA for the given ε-NFA — 3
DFA for the NFA found — 3



Equivalent NFA.



Equivalent DFA

3.a) i) Equivalent DFA for $(a+b)^* ab (a+b)^*$ — 2

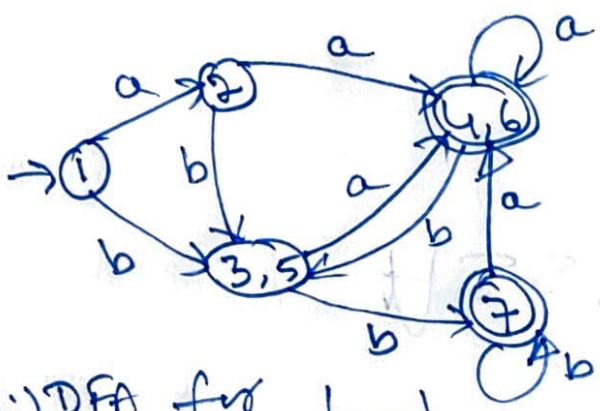
ii) Equivalent DFA for $(a+b)^* (ab + bba)$ — 2

3.b

	1	2	3	4	5	6
2	✓					
3	✓	✓				
4	✓	✓	✓			
5	✓	✓		✓		
6	✓	✓	✓		✓	
7	✓	✓	✓	✓	✓	✓

— 4

$3 \equiv 5$ & $4 \equiv 6$



— 2

4.a) i) DFA for $L_1 - L_2$ — 2

ii) DFA for $L_1 \cap L_3$ — 2

iii) DFA for $L_2 \cup L_3$ — 2

4.b) Decision algorithm for (i) — 2

Decision algorithm for (ii) — 2

5.a) pumping lemma statement — 1
proof of the lemma — 2
To show the given language is not regular — 3

5.b) a) NO. It won't work — 1
example — 1

b) NO. It won't work — 1
example. — 1